

The Thermic Degree of a Norton Product

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Abstract

Let $G.u$ denote Norton multiplication of a short game by a positive dyadic number $u = m/2^k$ in canonical form. Put $\delta_u = 1/\text{den}(u) = 2^{-k}$ and $a_u = u - \delta_u$. We prove the exact thermic law

$$\text{mean}(G.u) = u \text{mean}(G), \quad \text{temp}(G.u) = u \text{temp}(G) + a_u$$

for every nonnumeric game in the ordinary finite thermographic domain. For a number x , whose canonical mesh is $\epsilon_x = 1/\text{den}(x)$, the second formula is replaced by

$$\text{temp}(x.u) = \begin{cases} -1, & a_u - u\epsilon_x < 0, \\ a_u - u\epsilon_x, & a_u - u\epsilon_x \geq 0. \end{cases}$$

Thus every positive numeric Norton operator descends to an additive transport from thermic degree τ to degree $u\tau + a_u$.

These transports do not form a dyadic action. Their exact degree defect is

$$\Delta(u, v) = v(1 - \delta_u) - \delta_v + \delta_{uv} \geq 0,$$

and we classify all pairs for which it vanishes. The formula, its numeric branch, and the complete defect classification are the main results. We end by locating them inside the more modest tropical comparison: temperature is a filtration degree and thermographic option folds use max-plus and min-plus, but the freeze step is not tropical. Moreover the degree-zero residue is the group of infinitesimals and contains $[*] \neq 0$ with $2[*] = 0$, so no faithful lift-compatible action of full dyadic coefficients can turn the filtration into a Newton-style graded ring.

1 Norton and thermographic conventions

We work with short normal-play partizan games. Numbers are assigned temperature -1 ; every game discussed below is in the ordinary finite thermographic domain. The recursive definition and the standard order and additivity properties of Norton multiplication may be found in [4, Ch. 8], [7, §6.2], and [8, Prop. 13].

There is a form convention to make explicit. Norton multiplication is independent of the chosen form of its first argument, but its definition uses the selected Left and Right options of the positive unit [9]. Different presentations of the same unit can therefore give different recurrences: in the usual notation, for example, $(1/2).\{1* \mid\} = \{1 \mid 0\}$ whereas $(1/2).\{0 \mid\} = 1/2$. Throughout this paper a numeric unit means its *canonical short-game form*. This is also the convention of the shipped implementation, which canonicalizes the unit before extracting its incentives.

Let $u = m/2^k > 0$ be in lowest terms, and define

$$\delta_u = \frac{1}{\text{den}(u)} = 2^{-k}, \quad a_u = u - \delta_u. \tag{1}$$

For $k = 0$ this says $\delta_u = 1$. The canonical options of a nonintegral u are $u - \delta_u$ and $u + \delta_u$; a positive integer has the sole Left option $u - 1$. Every Norton increment is consequently the same number a_u : a Left option contributes $u - \delta_u$, and a Right option contributes $2u - (u + \delta_u)$. Writing $A_u(G) = G.u$, the defining recurrence reduces to

$$A_u(n) = nu \quad (n \in \mathbb{Z}), \quad A_u(G) = \{A_u(G^L) + a_u \mid A_u(G^R) - a_u\} \quad (G \notin \mathbb{Z}). \quad (2)$$

We next fix the thermographic notation used in the proof. For a number x , set $LS_t(x) = RS_t(x) = x$. For a nonnumber G , form the raw option envelopes

$$L_G^{\text{raw}}(t) = \max_{G^L} RS_t(G^L), \quad R_G^{\text{raw}}(t) = \min_{G^R} LS_t(G^R). \quad (3)$$

Before freezing, the left and right scaffold walls are

$$LS_t^{\text{sc}}(G) = L_G^{\text{raw}}(t) - t, \quad RS_t^{\text{sc}}(G) = R_G^{\text{raw}}(t) + t. \quad (4)$$

They are followed until their first meeting and are then replaced by the common constant mast. The resulting frozen walls are denoted $LS_t(G)$ and $RS_t(G)$; at $t = 0$ they give the ordinary stops $LS(G)$ and $RS(G)$. Thus

$$D_G(t) = L_G^{\text{raw}}(t) - R_G^{\text{raw}}(t) - 2t = LS_t^{\text{sc}}(G) - RS_t^{\text{sc}}(G), \quad (5)$$

and $\text{temp}(G)$ is the least $t \geq 0$ at which this raw gap closes. Wall slopes give that D_G is nonincreasing, LS_t is nonincreasing, and RS_t is nondecreasing.

For later use, if X is a Left option in (2), define its *transformed right wall at height s* by

$$\mathcal{T}_X^R(s) = RS_s(A_u(X)) + a_u - s. \quad (6)$$

This is exactly the candidate that X contributes to the left scaffold of $A_u(G)$. Dually, a Right option Y contributes the transformed left wall

$$\mathcal{T}_Y^L(s) = LS_s(A_u(Y)) - a_u + s.$$

The terminology will never refer merely to $RS_s(A_u(X))$ or $LS_s(A_u(Y))$; the parent shift in (6) is part of the definition.

We use number avoidance in its standard form: in a sum $H + x$ with x a number and H a nonnumber, any player with an optimal move has one in the H component. In particular, comparisons with a number can be proved by giving the option-wise strategy in the nonnumeric component [10, Part II]. We will also use the standard stop implication

$$RS(H) < m \implies H \not\geq m, \quad (7)$$

which says equivalently that Right wins $H - m$ when moving first; the mirror statement uses $LS(H) > m$.

2 The exact positive-dyadic law

The thermic formula appears not to have been located in the literature.¹ The proof needs a separate numeric branch because Norton recursion stops at integers, not at arbitrary numbers.

¹The electronically accessible accounts in [7, §6.2] and [8, Prop. 13] give Norton multiplication and its standard algebraic properties, but not this affine thermic law. Two print-only loci remain to be checked by hand before any stronger novelty claim is made: *Winning Ways*, Chapter 8, and Siegel, §III.3. Accordingly, “not located” is meant literally, not as a claim of absolute novelty.

Lemma 1 (numeric images). *Let x be a short-game number and put $\epsilon_x = 1/\text{den}(x)$. Then*

$$\text{mean}(A_u(x)) = ux, \quad \text{temp}(A_u(x)) = \begin{cases} -1, & a_u - u\epsilon_x < 0, \\ a_u - u\epsilon_x, & a_u - u\epsilon_x \geq 0. \end{cases}$$

In particular $\text{temp}(A_u(x)) < a_u$.

Proof. Induct on the exponent of the exact dyadic denominator of x . Integers are the base clause of (2), so their images are the numbers ux and have temperature -1 . If x is nonintegral with canonical mesh $\epsilon = \epsilon_x$, then

$$x = \{x - \epsilon \mid x + \epsilon\}.$$

The two option meshes are 2ϵ (unless an option is already integral), so their candidate temperatures are $a_u - 2u\epsilon < a_u - u\epsilon$. By induction the option images have masts $u(x - \epsilon)$ and $u(x + \epsilon)$ and freeze strictly before the height $c = a_u - u\epsilon$ whenever $c \geq 0$. At height c the two parent scaffold values are therefore

$$u(x - \epsilon) + a_u - c = ux, \quad u(x + \epsilon) - a_u + c = ux.$$

On the terminal interval immediately below c their gap is $2(c - s) > 0$. Since the raw gap is nonincreasing, it could not have closed earlier. Hence the first meeting is at c and its mast is ux .

If $c < 0$, every lower-denominator option image is a number. The two numeric bounds in (2) are $ux + c < ux < ux - c$; the simplicity rule selects the number ux . This is the temperature- -1 branch. In both cases the displayed temperature is strictly below a_u , since $u\epsilon > 0$. $\square$

The next identity is the algebraic center of the proof. It is stated for the raw gap because that is where the two parent shifts cancel exactly.

Lemma 2 (affine raw-gap transport). *Let G be nonintegral, and suppose that for every option X of G and every $t \geq 0$,*

$$LS_{a_u+ut}(A_u(X)) = uLS_t(X), \quad RS_{a_u+ut}(A_u(X)) = uRS_t(X).$$

Then

$$L_{A_u(G)}^{\text{raw}}(a_u + ut) = uL_G^{\text{raw}}(t) + a_u, \tag{8}$$

$$R_{A_u(G)}^{\text{raw}}(a_u + ut) = uR_G^{\text{raw}}(t) - a_u, \tag{9}$$

and, in particular,

$$\boxed{D_{A_u(G)}(a_u + ut) = uD_G(t)}. \tag{10}$$

Proof. By (2), a Left option of $A_u(G)$ is $A_u(G^L) + a_u$. Numeric translation shifts both of its walls by a_u . Therefore (3) and the induction hypothesis give

$$\max_{G^L}(RS_{a_u+ut}(A_u(G^L)) + a_u) = \max_{G^L}(uRS_t(G^L) + a_u) = uL_G^{\text{raw}}(t) + a_u.$$

The Right-option calculation is the same with minimum and the shift $-a_u$, giving (9). Substitute both identities and $s = a_u + ut$ into the definition (5):

$$\begin{aligned} D_{A_u(G)}(s) &= (uL_G^{\text{raw}}(t) + a_u) - (uR_G^{\text{raw}}(t) - a_u) - 2(a_u + ut) \\ &= u(L_G^{\text{raw}}(t) - R_G^{\text{raw}}(t) - 2t). \end{aligned}$$

This is (10). $\square$

At positive input temperature, (10) immediately excludes a meeting below a_u . Temperature zero is subtler: both sides of (10) vanish at $t = 0$, so one must rule out a terminal mast already present below a_u . The following strategy lemma does exactly that.

Lemma 3 (full-slope corridor). *Assume $a_u > 0$. Let X be a canonical nonnumber, let m be a number, and assume that every proper follower Z of X already satisfies the endpoint identities*

$$LS_{a_u}(A_u(Z)) = uLS(Z), \quad RS_{a_u}(A_u(Z)) = uRS(Z).$$

Suppose that for some $b < a_u$,

$$RS_s(A_u(X)) = um - a_u + s \quad (b \leq s \leq a_u). \quad (11)$$

Equivalently, the transformed wall $\mathcal{T}_X^R(s)$ is the constant um on that interval. Then Right wins $X - m$ moving first. The mirror statement holds when $LS_s(A_u(X)) = um + a_u - s$: Left wins $X - m$ moving first.

Proof. We prove the Right statement by induction on the birthday of X ; negation gives the mirror. Since the wall in (11) has the maximal allowed slope $+1$, it is an unfrozen right scaffold on $[b, a_u]$. Using (2) there gives

$$\min_{Y \in X^R} LS_s(A_u(Y)) = um \quad (b \leq s \leq a_u). \quad (12)$$

Choose a Right option Y attaining the minimum at $s = b$. Every left wall is nonincreasing, while (12) says that each such wall is at least um throughout the interval. Hence the chosen wall cannot fall:

$$LS_s(A_u(Y)) = um \quad (b \leq s \leq a_u). \quad (13)$$

If Y is numeric, Lemma 1 says it has already frozen strictly below a_u , and its mast at a_u is uY . Equation (13) therefore forces $Y = m$. Right moves from $X - m$ to $Y - m = 0$ and wins.

Suppose Y is nonnumeric. Let Z be any Left option of Y . Its transformed right wall is one of the candidates under the flat left wall (13); consequently

$$RS_s(A_u(Z)) + a_u - s \leq um \quad (b \leq s \leq a_u). \quad (14)$$

At $s = a_u$, the endpoint hypothesis gives $RS_{a_u}(A_u(Z)) = uRS(Z)$. Thus $RS(Z) \leq m$.

If $RS(Z) < m$, (7) says that Right wins $Z - m$ moving first. If $RS(Z) = m$, then the right wall's slope bound $+1$ and its endpoint um give

$$RS_s(A_u(Z)) \geq um - a_u + s.$$

Together with (14), this is equality on $[b, a_u]$. The induction hypothesis applied to the lower-birthday game Z again says that Right wins $Z - m$ moving first.

Right therefore begins $X - m$ by moving X to Y . After every Left reply $Y \rightarrow Z$, the preceding paragraph supplies Right's winning continuation from $Z - m$. Number avoidance says that Left gains nothing by moving in the numeric component $-m$ instead. This completes the explicit reply strategy and the birthday induction. $\square$

Theorem 1 (exact thermic law). *Let $u > 0$ be a dyadic number in canonical form, with δ_u and a_u as in (1). For every nonnumeric short game G in the finite thermographic domain,*

$$\text{mean}(A_u(G)) = u \text{mean}(G), \quad \text{temp}(A_u(G)) = u \text{temp}(G) + a_u. \quad (15)$$

More strongly, for every $t \geq 0$ the frozen walls obey

$$LS_{a_u+ut}(A_u(G)) = uLS_t(G), \quad RS_{a_u+ut}(A_u(G)) = uRS_t(G). \quad (16)$$

For numeric $G = x$, Lemma 1 is the exact replacement.

Proof. We prove (15) and (16) simultaneously by birthday induction. Numeric subpositions are supplied by Lemma 1; for nonnumeric options, use the birthday induction. Thus the hypotheses of Lemma 2 hold for G , and the raw-gap identity (10) is available.

Put $\tau = \text{temp}(G)$. If $\tau > 0$, then $D_G(0) > 0$, hence

$$D_{A_u(G)}(a_u) = uD_G(0) > 0.$$

Because the raw gap is nonincreasing, it cannot have closed at any smaller height. Equation (10) then says that its first zero at or above a_u occurs exactly at $a_u + u\tau$.

If $\tau = 0$ and $a_u = 0$, the same identity locates the first meeting at zero. It remains only to treat $\tau = 0 < a_u$. Here $LS(G) = RS(G) = m$ for a number m , and (10) locates a meeting of the transformed raw scaffolds at a_u with value um . Suppose for a contradiction that $A_u(G)$ had already frozen at some $b < a_u$. Its two walls would then be the constant mast um throughout $[b, a_u]$.

Consider an arbitrary Left option X of G . At $s = a_u$, the induction hypothesis gives

$$RS_{a_u}(A_u(X)) = uRS(X), \quad \max_{G^L} RS(X) = LS(G) = m.$$

If $RS(X) < m$, then Right wins $X - m$ moving first by (7). If $RS(X) = m$, the candidate $\mathcal{T}_X^R(s)$ lies below the frozen parent mast um on $[b, a_u]$ and equals it at a_u . Since $RS_s(A_u(X))$ has slope at most $+1$, $\mathcal{T}_X^R$ is nonincreasing; going backwards from its endpoint shows it cannot be strictly smaller. Therefore

$$\mathcal{T}_X^R(s) = um, \quad \text{or equivalently} \quad RS_s(A_u(X)) = um - a_u + s \quad (b \leq s \leq a_u).$$

The endpoint hypotheses of Lemma 3 are exactly the already proved birthday-induction statements for the proper followers of X ; the lemma therefore says that Right wins $X - m$ moving first.

The quantifier is the point: the argument has been applied to *every* Left option $X = G^L$. Hence every Left first move in the G component of $G - m$ loses to Right. By number avoidance, a move in the number component is not better. Right therefore wins $G - m$ playing second, which proves $G \leq m$. The mirror argument, applied option by option to every Right option G^R and the left-wall half of Lemma 3, proves $G \geq m$. Thus $G = m$, contrary to the hypothesis that G is nonnumeric. No premature meeting below a_u is possible.

We have proved in all cases that the first meeting is $s = a_u + u\tau$. At $s = a_u + ut$ before that meeting, (8) and (4) give

$$LS_s^{\text{sc}}(A_u(G)) = uL_G^{\text{raw}}(t) + a_u - (a_u + ut) = uLS_t^{\text{sc}}(G),$$

and the right-wall calculation is identical. At the meeting, their common value is $u \text{mean}(G)$; above it both sides freeze to that mast. This proves (16) and both formulas in (15). $\square$

Corollary 1 (thermic residue transport). *Let $F_{<0} = \mathcal{N}$ be the subgroup of short numbers and, for $\tau \geq 0$, let $F_{\leq \tau}$ be the subgroup of games of temperature at most τ , including $F_{<0}$. Then A_u induces an additive map*

$$\overline{A}_{u,\tau} : F_{\leq \tau} / F_{< \tau} \longrightarrow F_{\leq u\tau + a_u} / F_{< u\tau + a_u}. \quad (17)$$

It is the generalized overheating operator $\int_u^{a_u}$. For a positive integer n , the shifted thermal height $h = \tau + 1$ scales exactly: $h(A_n(G)) = nh(G)$ on nonnumeric layers.

Proof. Norton multiplication by a fixed positive unit is additive [8, Prop. 13]. If $G - H$ is nonnumeric of temperature $\sigma < \tau$, Theorem 1 gives $\text{temp}(A_u(G - H)) = u\sigma + a_u < u\tau + a_u$. If $G - H$ is numeric, Lemma 1 puts its image strictly below a_u , hence below the target layer. Thus (17) is independent of the representative. Recurrence (2) is precisely $\int_u^{a_u}$. Finally $a_n = n - 1$, so $\text{temp}(A_n(G)) + 1 = n(\text{temp}(G) + 1)$. $\square$

3 The exact composition defect

Write

$$r_u(\tau) = u\tau + a_u = u\tau + u - \delta_u.$$

Corollary 1 says that A_u carries thermic degree τ to $r_u(\tau)$. The discrepancy between successive transports and transport by the ordinary dyadic product is completely arithmetic.

Theorem 2 (composition defect). *For positive dyadics u, v and every $\tau \geq 0$,*

$$r_v(r_u(\tau)) = r_{uv}(\tau) + \Delta(u, v), \quad \Delta(u, v) = v(1 - \delta_u) - \delta_v + \delta_{uv} \geq 0. \quad (18)$$

The defect vanishes exactly in the following cases:

- (i) u is nonintegral and $v = 2^{-\ell}$ for some $\ell \geq 0$;
- (ii) u is an odd positive integer and v is arbitrary;
- (iii) u is an even positive integer and v is an integer.

Proof. Expanding the affine maps gives

$$\begin{aligned} r_v(r_u(\tau)) - r_{uv}(\tau) &= v(u\tau + u - \delta_u) + v - \delta_v - (uv\tau + uv - \delta_{uv}) \\ &= v(1 - \delta_u) - \delta_v + \delta_{uv}. \end{aligned}$$

If u is integral, $\delta_u = 1$ and $\Delta = \delta_{uv} - \delta_v$. Multiplication by an integer cannot increase a dyadic denominator. It preserves the denominator of every v exactly when u is odd; if u is even and v nonintegral, it cancels at least one factor of two. Integral v always has $\delta_{uv} = \delta_v = 1$. This gives (ii), (iii), and nonnegativity in the integral- u case.

Now write $u = m/2^k$ in lowest terms with $k > 0$. If v is nonintegral, both numerators are odd and $\delta_{uv} = \delta_u\delta_v$. Hence

$$\Delta = (1 - \delta_u)(v - \delta_v) \geq 0,$$

with equality exactly when v has numerator 1, that is, $v = 2^{-\ell}$. If $v = n$ is integral, write $n = 2^s q$ with q odd. For $s \geq k$, one has $\delta_{uv} = 1$ and $\Delta = n(1 - 2^{-k}) > 0$ unless $n = 1$. For $s < k$,

$$\Delta = n - 1 - \frac{n - 2^s}{2^k}.$$

This is zero at $n = 1$ and positive at every $n > 1$: for $s = 0$ it is $(n - 1)(1 - 2^{-k})$, while for $s > 0$ the subtracted fraction is at most $(n - 2^s)/2 < n - 1$. This proves (i) and completes the classification. $\square$

Corollary 2 (not a dyadic action). *No temperature-preserving residue refinement can make all numeric Norton transports into the scalar action of an associative algebra whose positive dyadic coefficients multiply ordinarily.*

Proof. Take $u = 1/2$, $v = 2$, and $G = *$. Theorem 2 gives $\Delta(1/2, 2) = 1$, hence

$$\text{temp}(A_2 A_{1/2}(*)) = 1, \quad \text{temp}(A_1(*)) = \text{temp}(*) = 0.$$

Associativity would identify the two outputs, but a refinement preserving exact thermic degree cannot identify classes in different degrees. $\square$

4 Machine verification in this repository

The proof above is mathematical; the implementation supplies independent regression oracles for its formulas. The following helpers live in `src/games/heating.rs`:

```
numeric_norton_regrade    numeric_norton_mean_temperature.
```

The first returns (u, a_u) ; the second evaluates both branches of Theorem 1 without materializing the recursive product. Source tests compare those helpers with actual Norton products on integers, nonintegral numbers, all-small games, switches, and a nested hot game, using eight units from $1/4$ through 3. A separately transcribed recursive Norton oracle checks the noninteger-game/noninteger-unit branch.

The maintained probe `experiments/under_descent.py` uses a 35-game catalogue and six units. It checks 210 exact mean/temperature predictions, 24 representative-pair descent tests for Norton multiplication, and 24 for the matching overheating operator. Its arithmetic defect scan checks 2,304 positive-dyadic pairs for nonnegativity and the complete zero classification, then compares five selected defects with recursively materialized game products. It also records the explicit nonnumeric-unit failures discussed below. These are bounded machine checks, not substitutes for the universal proofs.

5 The tropical shadow and its limits

5.1 What is genuinely tropical

Temperature satisfies the standard maximum inequality

$$\text{temp}(G + H) \leq \max(\text{temp}(G), \text{temp}(H)), \quad (19)$$

with numbers colder than every infinitesimal or hot game [10, Thm. II.5.18]; see also the cooling-homomorphism treatment in [3]. Consequently the $F_{\leq \tau}$ above are additive subgroups and define

$$\text{gr}_T(\mathcal{G}) = \bigoplus_{\tau \geq 0} F_{\leq \tau} / F_{< \tau}.$$

Equal leading temperatures may cancel to a lower layer, just as equal valuations may cancel in a valued field.

There is also an exact max-plus/min-plus observation inside the wall recursion: L_G^{raw} is the pointwise maximum of the Left options' right walls, and R_G^{raw} is the pointwise minimum of the Right options' left walls. Cooling shifts wall values by ordinary addition. These are dual tropical folds. The subsequent operation is not tropical, however: one subtracts the two envelopes and $2t$, finds the first zero of D_G , and freezes both walls to a common mast. The module `src/games/tropical_thermography.rs` deliberately reuses this ordinary freeze tail; it only names the two option folds. Thus thermography contains tropical operations but is not, as a whole, a semiring tropicalization.

The valuation-theoretic comparison remains useful. A Newton polygon takes the lower convex hull of $(i, v(a_i))$, and Dumas's theorem makes the polygon of a product the appropriate sum of the factor polygons [6]. The game side has the filtration inequality (19) and the external regradings (17), but no corresponding internal product on Conway game values.

Two terminological distinctions prevent predictable misreadings. First, “tropical games” often means the algorithmic equivalence between tropical polyhedra and finite mean-payoff games established by Akian–Gaubert–Guterman [1]. Those are weighted graph optimization games, not

Conway values or thermographs. Second, “temperature” in idempotent analysis commonly denotes the Maslov dequantization parameter whose zero limit turns log-sum-exp into max-plus arithmetic [11]. Combinatorial-game temperature instead measures the urgency at which cooling freezes a game. The shared words do not identify the two constructions.

5.2 Thermographs and nonnumeric units do not descend

The exact thermograph is not a congruence for disjunctive sum. Indeed, $*$ and $\uparrow$ have the same constant-zero walls, mast zero, and temperature zero. Yet $* + *$ has temperature -1 , whereas $\uparrow + *$ is a nonnumber of temperature zero. Hence no binary operation on thermographs alone can recover the thermograph of every sum.

Nor does arbitrary Norton multiplication descend to gr_T . At degree zero, $*$ and $* + 1$ differ by the cold number 1. For the positive nonnumeric unit $U = \uparrow$, additivity and the integer clause give the two-line obstruction

$$A_U(* + 1) - A_U(*) = A_U(1) = U = \uparrow.$$

The output difference has a nonzero temperature-zero class. Thus numericity of u in Corollary 1 is a real boundary, not a proof convenience.

5.3 The degree-zero torsion remark

The degree-zero group is already familiar. Every temperature-zero game is a number plus an infinitesimal, and quotienting by the cold numbers gives

$$\text{gr}_0(\mathcal{G}) = F_{\leq 0}/F_{< 0} \cong \mathcal{I}, \quad (20)$$

the additive group of infinitesimals. Calistrate’s decomposition $\mathcal{G} = \mathcal{I} \oplus \text{Rcf}$ places this group inside the classical structure theory [5]. Its torsion is not created by the temperature filtration: $[*] \neq 0$ and $2[*] = 0$ reflect the pre-existing additive group of games, studied abstractly by Moews [9]. Once (20) is stated, the coefficient obstruction is the elementary fact that inverting 2 kills nonzero 2-torsion. We record the precise contract only to prevent a grading loophole.

Definition 1 (full-dyadic lift-compatible action). Such an action consists of the following data.

- (i) An abelian grading group Γ , a degree map $\iota_T : \mathbb{Q}_{\geq 0} \rightarrow \Gamma$, and a Γ -graded additive group $R = \bigoplus_{\gamma \in \Gamma} R_\gamma$.
- (ii) Injective homomorphisms $j_\tau : F_{\leq \tau}/F_{< \tau} \hookrightarrow R_{\iota_T(\tau)}$ for every thermic degree τ .
- (iii) A graded unital ring $C = \bigoplus_d C_d$ acting associatively and distributively on R , with $C_d R_\gamma \subseteq R_{d+\gamma}$.
- (iv) Mutually inverse homogeneous elements $\pi_2 \in C_d$ and $\pi_{1/2} \in C_{-d}$ representing the full dyadic coefficients 2 and $1/2$.
- (v) A common Γ -filtered lift of the game group extending the thermic filtration, such that $j_\tau([G]_\tau)$ is the $\iota_T(\tau)$ -initial class of G and, in the single component $R_{\iota_T(\tau)+d}$,

$$\pi_2 j_\tau([G]_\tau) = \text{in}_{\iota_T(\tau)+d}(G + G). \quad (21)$$

The right side is zero when $G + G$ lies strictly below that common target degree, and in particular when $G + G = 0$.

The common group Γ and the component declaration in (21) are part of the data; neither side may be silently regraded into a different group.

Proposition 1 (dyadic torsion no-go). *No faithful full-dyadic lift-compatible action exists.*

Proof. Let $s = j_0([\ast]) \neq 0$. Since $\ast + \ast = 0$, compatibility gives $\pi_2 s = 0$. But π_2 is invertible, so

$$s = (\pi_{1/2}\pi_2)s = \pi_{1/2}(\pi_2 s) = 0,$$

a contradiction. □

This does not forbid three alternatives: integer or valuation-ring coefficients in which 2 is not inverted; a characteristic-2 quotient that kills the incompatible coefficient data; or a ring built from a finer equivalence relation rather than Conway equality. The recent construction of Altman–Lipparini belongs to the last category [2]. It is adjacent to, but does not contradict, Proposition 1.

6 Conclusion

The positive theorem is the durable bridge. A canonical positive dyadic unit u sends every non-numeric thermic degree by the exact affine rule $\tau \mapsto u\tau + u - \delta_u$, while numeric inputs occupy a strictly lower branch. This yields honest additive maps between thermic residue groups. The composition defect then gives the exact limit: these maps form a family of external regratings, not an action of the multiplicative dyadics.

The Newton-polygon comparison should be kept at that scale. Temperature obeys a nonarchimedean maximum inequality; the raw thermograph recursion contains dual tropical folds. The freeze operation is not tropical, the complete thermograph is not a sum congruence, and the degree-zero residual group retains the old infinitesimal torsion. The game and place sides therefore share a filtered shadow, not one faithful full-dyadic graded ring.

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